Particle Swarm Optimization

How individual memory and social information balance exploration and exploitation

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Why particle swarms?

Member 1

- Minimise a function with no usable gradient
- Local search stops at the first basin
- PSO: many particles that share what they find

Guiding question

How does PSO combine individual experience and social information to balance exploration and exploitation?

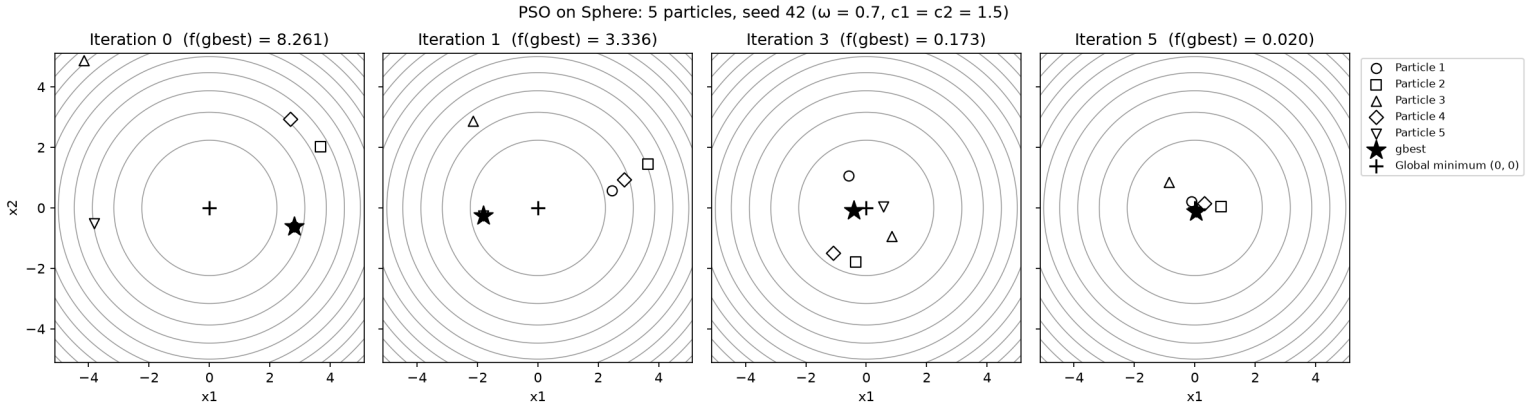


Figure: five particles converging on Sphere (team experiment)

From flocking to an optimiser

Member 1

$$v \leftarrow v + 2 \cdot \text{rand}() \cdot (\text{pbest} - x) + 2 \cdot \text{rand}() \cdot (\text{gbest} - x)$$

1. **pbest**
   Each agent remembers its own best position ("simple nostalgia")
2. **gbest**
   Best position known to everyone (a group norm)
3. **× 2**
   Mean pull of 1, so agents overshoot about half the time

Unbalanced pulls fail: a much stronger pull to gbest made the flock rush prematurely towards local minima; a stronger pull to pbest made agents wander. Removing the old velocity was "quite ineffective at finding global optima".

Kennedy & Eberhart (1995), pp. 1945–1946

What changed by 2007

Member 1

	Kennedy & Eberhart (1995)	Poli et al. (2007)
Velocity control	Fixed coefficient 2, no damping	Inertia ω (1 − ω acts as friction) or constriction χ
Recommended setting	Not studied	$\chi \approx 0.7298 \Leftrightarrow \omega = 0.7298$, $c_1 = c_2 \approx 1.496$
Who informs a particle	Whole swarm (gbest)	gbest is fast; lbest ring resists local optima
Evidence	A few demonstrations	About 700 application papers reviewed

Our default: $\omega = 0.7$, $c_1 = c_2 = 1.5$

A rounded version of the standard constriction setting. High ω explores, low ω exploits; we test 0.4, 0.7 and 0.9.

Poli, Kennedy & Blackwell (2007), Sects. 2.2–2.4, 3.1, 6

Problem, functions and update rule

Member 2

Sphere

$f(x) = \Sigma\, x_i^2$

One basin: tests convergence speed

Rastrigin

$f(x) = 20 + \Sigma\, (x_i^2 - 10 \cos 2\pi x_i)$

Grid of local minima: tests trapping

Both in $[-5.12, 5.12]^2$, global minimum 0 at (0, 0); fitness = f(x), lower is better

$$v \leftarrow \omega\, v + c_1 r_1 (pbest - x) + c_2 r_2 (gbest - x), \quad x \leftarrow x + v$$

inertia · cognitive: own best · social: swarm best

2,020

evaluations per run: 20 particles × 100 iterations + 20 at start; $|v| \le 2$; r_1, r_2 new for every particle, dimension and iteration

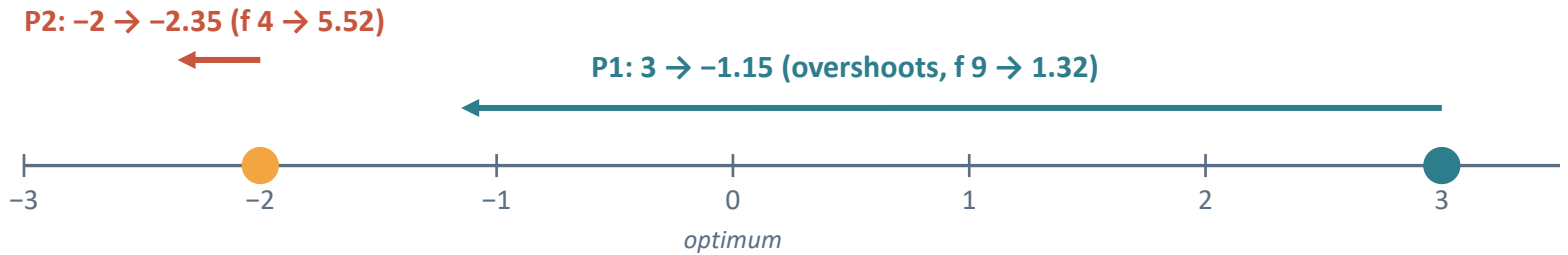
Update rule: Poli et al. (2007), Eq. 2. Functions and settings: team design

One step by hand

Member 2

$f(x) = x^2$, $\omega = 0.7$, $c_1 = c_2 = 1.5$; start: $x_1 = 3$, $x_2 = -2$ (gbest)

Particle	Inertia	Cognitive	Social	v(1)	x(1)	f(x(1))
1	0.35	0	1.5·0.6·(−5) = −4.5	**−4.15**	**−1.15**	1.3225
2 (gbest)	−0.35	0	0	−0.35	−2.35	5.5225

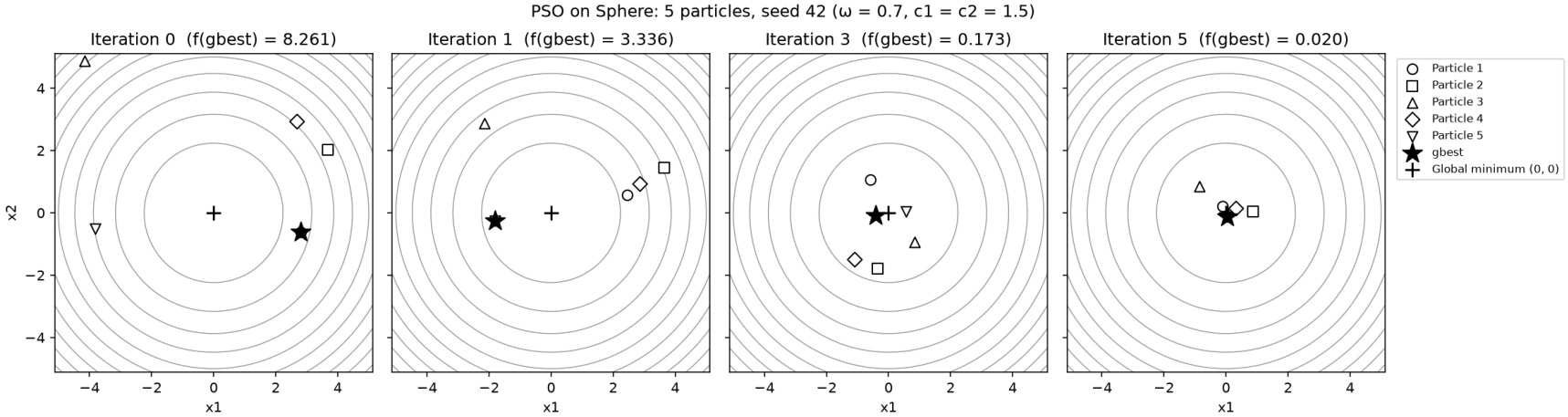


Cognitive term is 0 at the first step (pbest = x). One particle gets worse, yet gbest improves from −2 to −1.15.

Team worked example (report Table 3.2)

Five particles, five iterations

Member 2



f(gbest) by iteration

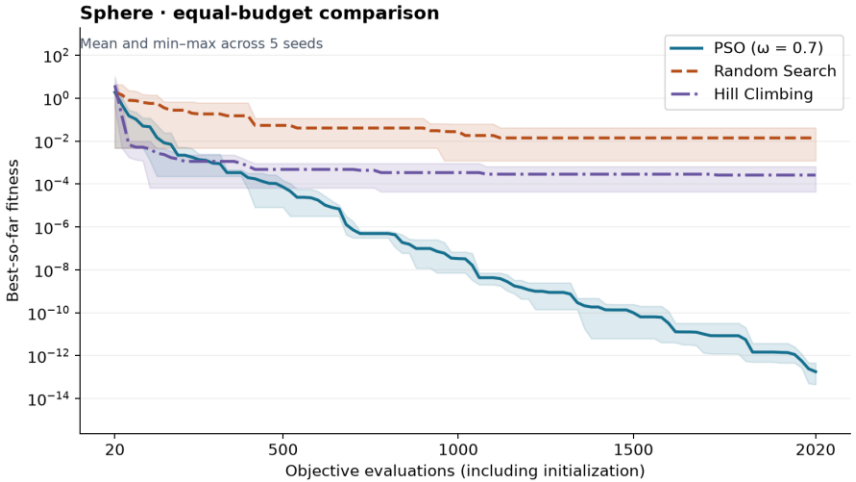
iter 0	iter 1	iter 2	iter 5
8.26	**3.34**	**0.17**	**0.02**

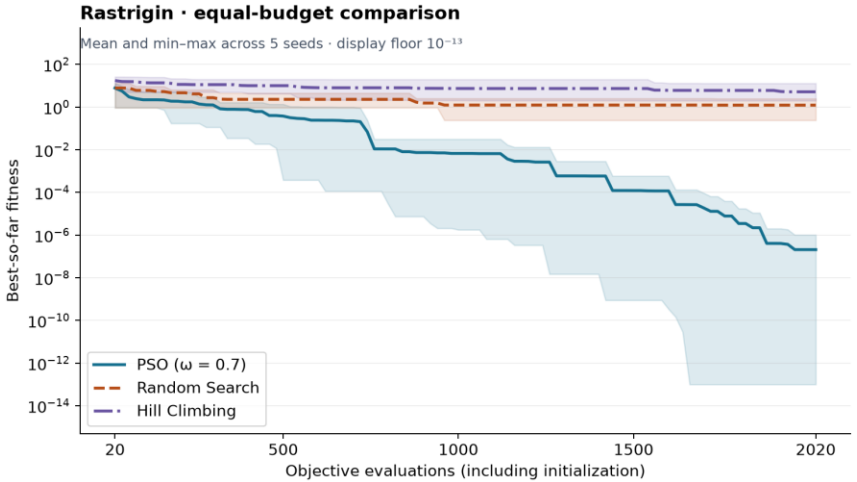
- Iteration 0 → 1: every particle improves
- Velocity clamp (±2) active for 3 particles
- Code reproduces the hand recomputation exactly

Team experiment: Sphere, 5 particles, seed 42 (report Table 3.3, Figure 3.1)

Same budget, two landscapes

Member 3





42.6 vs 132
evaluations to reach 0.01 on Sphere: Hill Climbing first

1.8×10^{-13}
PSO mean final fitness on Sphere, best of the three

5 / 5 vs 0 / 5
Rastrigin runs reaching 0.01: PSO vs each baseline

Team experiments: 2,020 evaluations per run, seeds 0–4 (report Tables 4.1–4.2, Figure 4.1)

Inertia depends on the landscape

Member 3

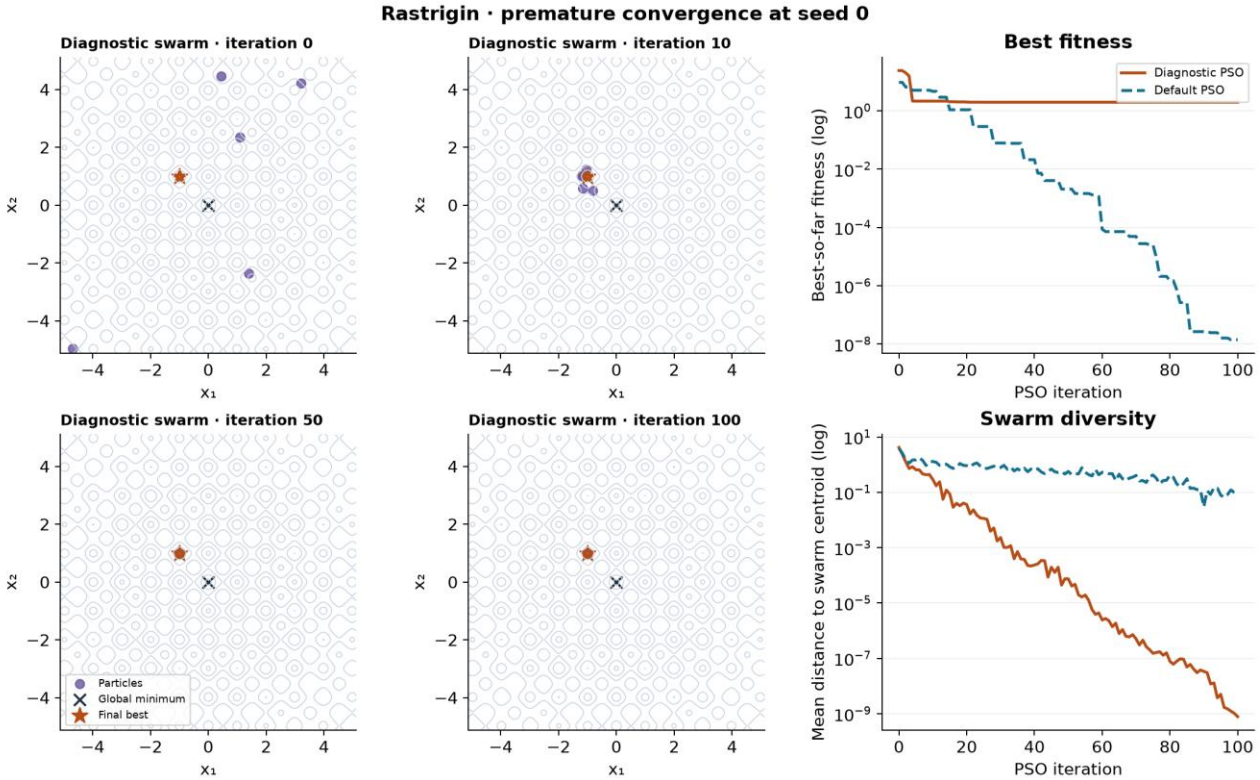
Mean final fitness	A: ω = 0.4	B: ω = 0.7	C: ω = 0.9
Sphere	**5.5×10^{-31}**	1.8×10^{-13}	4.2×10^{-5}
Rastrigin	0.199	**2.1×10^{-7}**	8.5×10^{-3}

1. **Sphere:** One basin: strong damping (low ω) settles and refines fastest
2. **Rastrigin:** Four ω = 0.4 runs reach ≈ 0; one stops at 0.995 and dominates the mean
3. **Velocity limit:** With v_max ≥ 10.24 (domain width) the stalled ω = 0.4 run disappears

Team experiment: 20 particles, $c_1 = c_2 = 1.5$, v_max = 2, seeds 0–4 (report Table 4.3, v_max check in Sect. 4.2)

Where the swarm got stuck

Member 3



Setting

5 particles, $\omega = 0.4$, $c_1 = 0.5$, $c_2 = 2.5$

Result

Local minimum at (−0.995, 0.995), $f \approx 1.99$

Diversity

Mean distance to centroid $4.18 \rightarrow 7.9 \times 10^{-10}$

Default run

Same seed stays spread out, $f \approx 1.4 \times 10^{-8}$

Team experiment: Rastrigin, seed 0 (report Figure 4.2)

Exploration and exploitation

Member 3

1

Personal best
keeps each particle's discoveries and the swarm's diversity

2

Global best
focuses the swarm on the best region found so far

3

Inertia
decides how far particles carry on past what attracts them

Too much exploitation (low inertia, strong social pull, few particles) can collapse the swarm onto a suboptimal basin.

Applications: image analysis, power systems and control lead PSO use (Poli et al., 2007), e.g. solar power tracking (Alshareef et al., 2019) and UAV path planning (Phung & Ha, 2021).
Limits: two 2-D functions with the optimum at the origin, five seeds, one topology.

Thank you · Questions?

Kennedy, J. & Eberhart, R. (1995). Particle swarm optimization. Proc. ICNN'95, 1942–1948.

Poli, R., Kennedy, J. & Blackwell, T. (2007). Particle swarm optimization: An overview. Swarm Intelligence 1, 33–57.

Alshareef, M. et al. (2019). PSO for PV power point tracking. Energies 12(4), 623.

Phung & Ha (2021). Spherical vector-based PSO for UAV path planning. Appl. Soft Comput. 107, 107376.

AI tools (Claude, OpenAI Codex) assisted with code and drafting; all results were checked by the team.